

(a) Data Collection



(b) 3D Geometry-Constrained QA Generation

Pass 1:

Object [0]

Pass 2:

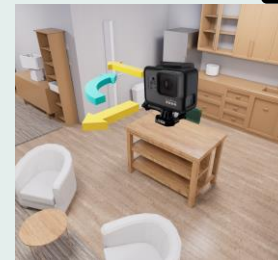
```
{  
  "cabinet": {"count": 7,  
    "ids": [0, ..., 26],  
    "instances":  
      ["cabinet_221", "c..."],  
    "table": {"count": 6,  
      "ids": [1, ..., 18],  
      "instances":  
        ["table_204", "tab..."],  
      ...  
}
```

```
{  
  "id": 0, "classname":  
    "cabinet", ...  
  // 2D BBox(px)  
  "bdb2d": { ... },  
  // Pano Bearing-  
  FOV  
  "bfov": { ... },  
  // 3D Bbox  
  "bdb3d": { ... }  
}
```

3D Mathematical and Geometric Relations



Relative Distance



Egocentric Rotation

QA Pairs

(c) Quality Assurance and Validation

Layer 1:
Auto Scripts
GT Audit

✓ Logical Error
✓ Numerical
✓ Out-of-Bounds

dump
X Numerical

Layer 2: Human in the loop

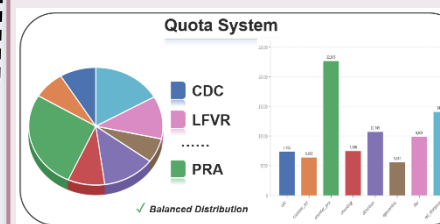


✓ Overly Convoluted
✓ Phrasing ✓ 3D layout

dump
X Out-of-Bounds

Dataset.json

(d) Dataset Partitioning



Total QA Pairs: 84,373
Panorama Images: 2,600
Indoor Environments: 26
Dataset | Partition



Train Sets



Test Sets

Canonical 3D Scene Layout Construction



Manu
Check

Raw omnidirectional image with annotations

2D bbox & 2D location

2D OmniDirectional Image bounding box



3D bbox &
3D location